3.2.2

EE25BTECH11061 - V.Sainadh

Question:
Draw a triangle $\triangle ABC$ in which $AB = 5\,\text{cm}$, $BC = 6\,\text{cm}$ and $\angle ABC = 60°$.

Solution:
→ Let

$$a = \|\mathbf{C} - \mathbf{B}\| = 6\,\text{cm} \tag{1}$$
$$b = \|\mathbf{A} - \mathbf{C}\| \quad \text{(to be found)} \tag{2}$$
$$c = \|\mathbf{B} - \mathbf{A}\| = 5\,\text{cm} \tag{3}$$

→ By cosine law in $\triangle ABC$ at **B**,

$$b^2 = a^2 + c^2 - 2ac\cos B \tag{4}$$
$$= 6^2 + 5^2 - 2 \cdot 6 \cdot 5 \cdot \cos 60° \tag{5}$$
$$= 36 + 25 - 60 \cdot \frac{1}{2} \tag{6}$$
$$= 31 \quad \Rightarrow \quad b = \sqrt{31} \approx 5.57\,\text{cm} \tag{7}$$

→ Choose coordinates (placing B at the origin and BC on the x-axis):

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{8}$$
$$\mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{9}$$
$$\mathbf{A} = c\begin{pmatrix} \cos B \\ \sin B \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{5\sqrt{3}}{2} \end{pmatrix} \approx \begin{pmatrix} 2.5 \\ 4.33 \end{pmatrix} \tag{10}$$

→ Therefore, the vertices of $\triangle ABC$ are:

$$\mathbf{A} = \begin{pmatrix} 2.5 \\ 4.33 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{11}$$

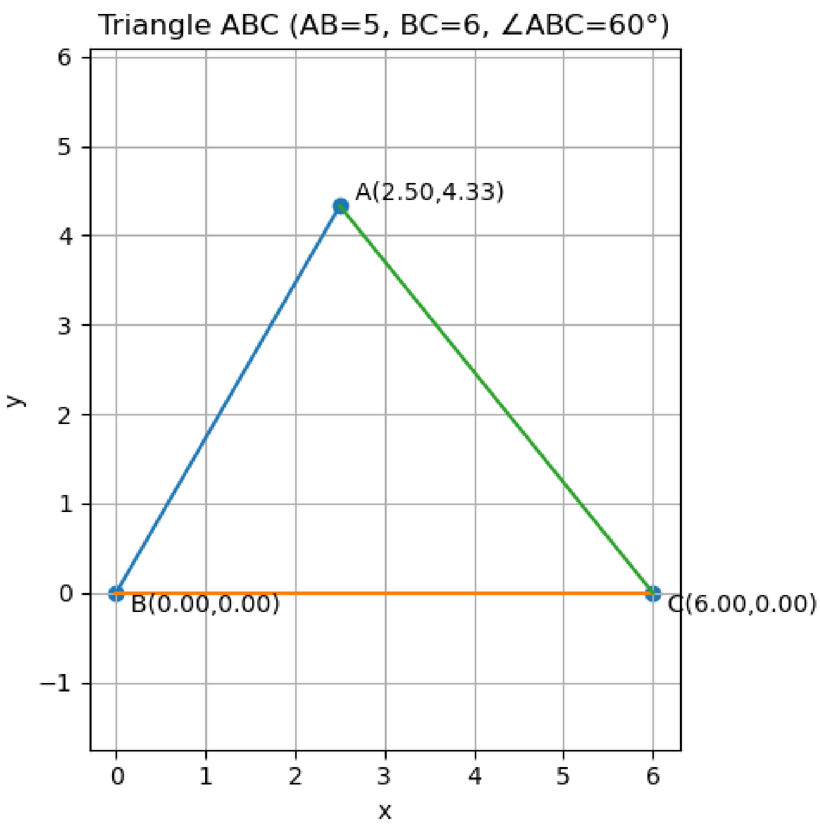

Fig. 0: Plot of $\triangle ABC$